

Computational Intelligence (CI)

Computational Intelligence is a branch of Artificial Intelligence that develops intelligent systems capable of learning, adapting, and solving complex real-world problems.

It includes techniques such as: Neural Networks, Fuzzy Logic, Evolutionary Computing, Swarm Intelligence

Natural Language Processing (NLP)

Natural Language Processing (NLP) is a field of AI that enables computers to understand, process, interpret, and generate human language.

NLP combines: Artificial Intelligence, Machine Learning, Computational Linguistics

Word Embedding Techniques

Word Embedding is a method of converting text into numerical vectors so that machine learning models can understand and process language.

Word embedding converts words into mathematical representations while preserving semantic meaning.

Example:

King $\rightarrow$ [0.25, 0.78, 0.12]
Queen $\rightarrow$ [0.30, 0.82, 0.10]

Words with similar meanings have similar vector representations.

Types of Word Embedding Techniques

1. Bag of Words (BoW)
2. TF-IDF
3. Word2Vec
4. GloVe
5. Neural Word Embedding

1. Bag of Words (BoW)

Bag of Words is a text representation technique that converts text into numerical vectors based on word occurrence frequency.

It ignores:

- Grammar
- Word order
- Context

Theory / Working

Step 1:

Create vocabulary from all unique words.

Step 2:

Count occurrences of each word.

Step 3:

Represent document as vector.

Example

Sentences:

I love AI
I love NLP

Vocabulary:

[I, Love, AI, NLP]

Vectors:

Sentence 1 $\rightarrow$ [1,1,1,0]
Sentence 2 $\rightarrow$ [1,1,0,1]

Advantages

1. Simple implementation.
2. Easy vector creation.
3. Works well for small datasets.

Disadvantages

1. Ignores context.
2. High dimensionality.
3. Does not capture semantics.

Applications

- Text classification
- Spam detection
- Document analysis

2. TF-IDF (Term Frequency – Inverse Document Frequency) -
TF-IDF is a statistical technique used to measure the importance of a word in a document relative to a collection of documents.

TF-IDF gives:

- Higher weight to important words
- Lower weight to common words

Components

A. Term Frequency (TF)

Measures how often a word appears.

$$TF = \frac{\text{Word Count}}{\text{Total Words}}$$

B. Inverse Document Frequency (IDF)

Measures uniqueness.

$$IDF = \log \left(\frac{\text{Total Documents}}{\text{Documents containing word}} \right)$$

Formula

$$TF-IDF = TF \times IDF$$

Example

Word: "Machine"

Appears frequently in one document but rarely overall.

$\rightarrow$ High TF-IDF score

Advantages

1. Highlights important words.

2. Reduces effect of common words.
3. Better than BoW.

Disadvantages

1. Ignores word meaning.
2. Cannot capture context.

Applications

- Search engines
- Information retrieval
- Text ranking

3. Word2Vec

Word2Vec is a neural-network-based word embedding technique that converts words into dense vectors while preserving semantic relationships.

Developed by: Google

Word2Vec learns word relationships from surrounding words. Words used in similar contexts obtain similar vectors.

Architectures

A. CBOW (Continuous Bag of Words)

Predicts current word from neighbouring words.

Example: I ____ AI

Predict → love

B. Skip-Gram

Predicts surrounding words from current word.

Example: love → I, AI

Example

King – Man + Woman ≈ Queen

Advantages

1. Captures semantic meaning.
2. Dense representation.
3. Efficient for large datasets.

Disadvantages

1. Requires training data.
2. Context-independent.

Applications

- Recommendation systems
- NLP models
- Search systems

4. GloVe (Global Vectors)

GloVe is a word embedding technique that combines global statistical information with vector representation.

Developed by: Stanford University

GloVe learns word vectors using word co-occurrence statistics. Words appearing together frequently receive similar vectors.

Working

1. Create co-occurrence matrix.
2. Calculate probabilities.
3. Generate vector representations.

Example

Words:

King → Queen

India → Delhi

Related words appear closer.

Advantages

1. Captures global relationships.
2. Efficient learning.
3. Better semantic understanding.

Disadvantages

1. Requires large memory.
2. Static embeddings.

Applications

- NLP
- Chatbots
- Language modelling

5. Neural Word Embedding

Neural Word Embedding refers to techniques that use neural networks to generate meaningful vector representations of words.

Unlike traditional encoding methods:

- Words become dense vectors
- Similar words remain close
- Semantic relationships are preserved

Examples:

- Word2Vec
- GloVe

Working

1. Text Input
2. Neural Network
3. Vector Learning
4. Word Embedding Output

Example : Domestic Animals

Cat → [0.22,0.81]

Dog → [0.25,0.78]

Vectors are close because meanings are similar.

Advantages

1. Captures context and semantics.
2. Efficient learning.
3. Improves NLP performance.

Disadvantages

1. Computationally expensive.
2. Requires large datasets.

Applications

- Machine translation
- Sentiment analysis

Compare and Contrast Word2Vec and GloVe

Word2Vec and **GloVe (Global Vectors for Word Representation)** are popular word embedding techniques used in NLP. Both convert words into dense numerical vectors that capture semantic relationships between words, but they differ in how embeddings are learned.

Parameter	Word2Vec	GloVe
Full Form	Word to Vector	Global Vectors for Word Representation
Developed By	Google	Stanford University
Approach	Predictive Model	Count-Based Model
Training Method	Learns by predicting surrounding words	Learns using word co-occurrence statistics
Context Usage	Uses local context window	Uses global corpus statistics
Data Representation	Neural network-based	Co-occurrence matrix-based
Computational Complexity	Faster for large datasets	Requires building co-occurrence matrix
Semantic Relationships	Captures contextual relationships well	Captures global word relationships well
Examples	CBOW, Skip-Gram	GloVe Algorithm

1. Neural Machine Translation (NMT)

Neural Machine Translation (NMT) is a deep learning-based approach used to automatically translate text from one language to another using neural networks.

Unlike traditional translation systems, NMT translates complete sentences while preserving meaning and context.

Traditional machine translation translated word-by-word or phrase-by-phrase.

Neural Machine Translation:

- Understands sentence context
- Generates natural translations
- Uses encoder-decoder architecture

Working of Neural Machine Translation

Step 1: Input sentence is provided.

Step 2: Encoder converts text into numerical representation.

Step 3: Decoder generates translated sentence.

Step 4: Output sentence is produced.

Example

Input: I love Artificial Intelligence

Output: Ich liebe künstliche Intelligenz

Advantages

1. Produces natural translations.
2. Captures sentence context.
3. Handles long sequences.

Disadvantages

1. Requires large datasets.
2. Computationally expensive.

Applications

- Language translation
- Multilingual chatbots
- International communication

2. Seq2Seq (Sequence-to-Sequence) and NMT

Seq2Seq (Sequence-to-Sequence) is a neural network architecture used to convert one sequence into another sequence. It is widely used in Neural Machine Translation.

Seq2Seq converts: Input Sequence -> Output Sequence

Input and output lengths may differ.

Architecture of Seq2Seq

1. Encoder - Reads input sequence and creates context vector.

2. Context Vector - Stores learned information.

3. Decoder - Generates translated output.

Working

1. Input Sentence
2. Encoder
3. Context Vector
4. Decoder
5. Translated Sentence

Example

Input: How are you

Output: Wie geht es dir

Advantages

1. Supports variable length sequences.
2. Handles language translation effectively.
3. Learns contextual relationships.

Disadvantages

1. Long sentences may reduce performance.
2. Training requires large datasets.

Applications

- Translation

- Chatbots
- Text summarization
- Speech processing

3. Translation Metrics

Translation Metrics are evaluation measures used to determine translation quality. They compare machine-generated translation with reference translation.

Main metrics:

1. BLEU Score
2. BERT Score

A. BLEU Score (Bilingual Evaluation Understudy)

BLEU Score evaluates translation quality by comparing generated text with reference text using word overlap.

BLEU measures:

- Precision of words
- Matching phrases
- Sentence similarity

Score range:

0 → Poor Translation

1 → Perfect Translation

Formula - $BLEU = Precision \times Brevity\ Penalty$

Example

Reference: I love AI

Generated: I love Artificial Intelligence

Higher overlap → Higher BLEU

Advantages

1. Fast evaluation.
2. Easy comparison.

Disadvantages

1. Ignores semantic meaning.
2. Sensitive to wording.

B. BERT Score

BERT Score evaluates translation quality using contextual embeddings instead of exact word matching.

Based on: BERT

BERT Score:

- Measures semantic similarity
- Understands context
- Handles synonyms

Working

Reference Sentence -> Generate Embeddings -> Compare Semantic Similarity -> Calculate Score

Example

Reference: The boy is happy

Prediction: The child is joyful

BERT Score → High

BLEU → Lower

Advantages

1. Captures meaning.
2. More accurate evaluation.

Disadvantages

1. Computationally expensive.

Applications

- Translation evaluation
- NLP benchmarking

Traditional Metrics vs Neural Metrics for Machine Translation Evaluation

Parameter	Traditional Metrics	Neural Metrics
Evaluation Basis	Word and phrase matching	Semantic and contextual meaning
Examples	BLEU, ROUGE, METEOR	BERTScore, COMET, BLEURT
Context Understanding	Limited context understanding	Deep contextual understanding
Semantic Similarity	Poor at capturing meaning	Good at capturing meaning
Correlation with Human Judgment	Moderate	High
Computational Cost	Low	High
Accuracy	Lower	Higher
Handling Synonyms	Usually treats synonyms as different words	Recognizes similar meanings and synonyms
Implementation Complexity	Simple and easy to implement	More complex, requires neural models

5. Neural Style Transfer (NST)

Neural Style Transfer is a deep learning technique used to combine the content of one image with the artistic style of another image.

NST generates a new image by preserving:

- Content from one image
- Style from another image

Working of Neural Style Transfer (NST)

Step 1: Input Images

Two images are provided:

- **Content Image** → Contains objects and structure to preserve.
- **Style Image** → Contains artistic style, colors, and textures.

Step 2: Feature Extraction

A pre-trained CNN (e.g., **VGG-19**) extracts features from both images.

- Lower layers extract edges and textures.
- Higher layers extract content and objects.

Step 3: Content Representation

The network captures important content features such as shapes, objects, and layout from the content image.

Step 4: Style Representation

The network extracts style features such as colors, textures, brush strokes, and artistic patterns from the style image.

Step 5: Image Generation

A random image is initialized and gradually modified to match the content of the content image and the style of the style image.

Step 6: Loss Calculation

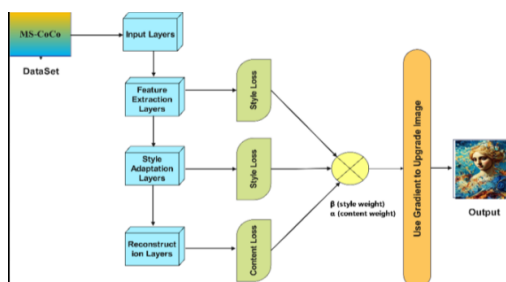
Two losses are computed:

- **Content Loss** → Difference between generated and content image.
- **Style Loss** → Difference between generated and style image.

Total Loss = Content Loss + Style Loss

Step 7: Optimization

The generated image is repeatedly updated to minimize total loss. The final image preserves the content of the first image and the artistic style of the second image.



Example

Content: Photo of city

Style: Painting style

Output: City image with painting appearance.

Advantages

1. Produces creative outputs.
2. Preserves content and style.

Disadvantages

1. Computationally expensive.

2. Slow generation.

Applications

1. Digital Art

Neural Style Transfer converts ordinary images into artistic paintings using the style of famous artworks.

2. Design

It helps designers create unique visual styles for products, websites, and graphic designs.

3. Entertainment

It is used in movies, games, and animations to generate creative visual effects and artistic scenes.

4. Media Editing

It enhances photos and videos by applying different artistic styles and visual transformations.

Pretrained NLP BERT Model and Applications

BERT (Bidirectional Encoder Representations from Transformers) is a pretrained language model designed to understand text context from both directions. Developed by: Google

Significance of Pre-trained BERT Models

1. Better Context Understanding

BERT reads text in both directions, allowing it to understand the complete context of words.

2. High Accuracy

It provides better performance than traditional NLP models in many language tasks.

3. Reduced Training Time

Since BERT is already pre-trained, only fine-tuning is required for specific applications.

4. Transfer Learning

The same pre-trained model can be adapted to multiple NLP tasks.

5. Handles Ambiguous Words

BERT understands word meaning based on context, improving language understanding.

Working of BERT

1. Bidirectional Learning

BERT reads text in both directions (left-to-right and right-to-left) simultaneously. This helps it understand the complete context of a word.

2. Transformer Encoder

BERT uses only the **Encoder** part of the Transformer architecture to process text and learn relationships between words.

3. Pre-Training

BERT is trained on large amounts of text using two tasks:

a) Masked Language Model (MLM)

- Some words are hidden (masked).

- BERT predicts the missing words.

Example:

I love [MASK] learning.

Output:

I love machine learning.

b) Next Sentence Prediction (NSP)

- Determines whether one sentence logically follows another.

4. Fine-Tuning

After pre-training, BERT can be fine-tuned for specific tasks such as sentiment analysis, text classification, question answering, and machine translation.

Features of BERT

- Bidirectional learning
- Context awareness
- Transfer learning
- Pretrained architecture

Parameter	Traditional Language Models	BERT
Context Understanding	One-directional	Bidirectional
Meaning Extraction	Limited	Better semantic understanding
Handling Ambiguous Words	Difficult	More accurate
Feature Engineering	Often required	Minimal requirement
Accuracy	Lower	Higher
NLP Tasks	Task-specific mod 	One model for multiple tasks

Applications of BERT

1. Question Answering

BERT can understand questions and accurately extract relevant answers from a given text.

2. Sentiment Analysis

BERT identifies the sentiment of text as positive, negative, or neutral.

3. Machine Translation

BERT helps improve translation quality by understanding the context and meaning of words.

4. Chatbots

BERT enables chatbots to understand user queries and generate more relevant responses.

5. Text Summarization

BERT helps create concise summaries while preserving the important information from a document.

6. Search Engines

BERT improves search results by understanding the intent and context of user queries.